

Zach Taylor

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SKILLS

Languages: C++, Python, Golang, Assembly, Typescript, JavaScript, HTML, CSS, SQL, Verilog

Libraries/Frameworks: MFC, React, Node.js, Next.js, Tailwind, TensorFlow, CUDA, FastAPI, MySQL

Developer Tools: Git, Docker, Linux, Visual Studio, Jenkins, Google Test, AWS, Vercel, SIMD, Microcontrollers

EXPERIENCE

Software Engineer Intern *Rohde & Schwarz*

April 2024 – Jan 2025

- Enabled advanced **network analysis** for the U.S. Army, Tesla, and first responders by delivering custom scanning features and frequency band configurations designed, implemented, and integrated using **C++**.
- Achieved a **25%** improvement in **processing speed** and maintainability by redesigning programs with efficient lookup algorithms and modern library features.
- Ensured stability and reliability across multiple software components by uncovering edge cases through rigorous **unit testing** and debugging with **Google Test**.
- Led meetings and cross-functional collaboration to support team alignment and ensure compliance with **agile methodologies**.

Software Development Intern *Portland State University*

June 2019 – September 2019

- Achieved **97%** object detection accuracy by implementing a **TensorFlow deep learning** model with **Python** and **OpenCV**, leveraging transfer learning for faster convergence and higher precision.
- Developed a versatile computer vision pipeline capable of processing both real-time camera streams and static datasets, enabling seamless deployment for online inference and batch processing.
- Led model integration, preprocessing, and postprocessing efforts, ensuring optimized inference speed and adaptability across diverse use cases.

Undergraduate Teaching Assistant *Oregon State University*

September 2020 – June 2023

- Strengthened students' foundational and practical skills by teaching **data structures** and **operating systems** in C, and guiding full stack web development using **JavaScript**, **Node.js**, and **HTML/CSS**.
- Raised academic performance by **30%** among struggling students by delivering personalized one-on-one and small group tutoring focused on reinforcing core concepts and addressing individual challenges.
- Improved students' **problem solving** abilities and debugging proficiency by walking through real code issues and encouraging a deeper understanding of troubleshooting techniques.

PROJECTS

Patient Monitoring Robot | *Python, OpenCV, ROS2, AI, Machine Learning*

[GitHub](#)

- Developed a robotic **AI** based fall detection system achieving **96%** accuracy using **Python**, **OpenCV**, and **machine learning** algorithms.
- Improved real time inference speed by **56%** through restructuring the **ROS2** based perception stack and streamlining **vision model** execution, reducing end-to-end latency and enhancing system responsiveness.
- Conducted field validation and **edge case testing** to ensure model robustness in variable lighting, occlusion, and movement patterns.
- Directed development efforts by mentoring team members on **AI model integration** and performance optimization, ensuring alignment with project requirements and milestones.

SIMD Image Filter | *C++, SIMD*

[Full Site](#)

- Designed and implemented a high performance image filtering library in **C++**, featuring blur, sharpen, and edge detection algorithms.
- Optimized image processing routines using **SIMD intrinsics**, resulting in **5x** faster execution compared to scalar implementations.

EDUCATION

Oregon State University

September 2019 – June 2023

Bachelor of Science in Computer Science

Corvallis, OR